Justin (Changjin) He

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References available on request

Skills

- **Programming:** Python (Pandas), TypeScript/React, C++, C, SQL (Vertica, DuckDB), R
- **Software Delivery:** FastAPI, Docker, Azure, Git, CI/CD workflows, Agile delivery, structured testing and documentation, AI-assisted development (Claude Code, GitHub Copilot, Cursor, ChatGPT)
- **Communication & Collaboration:** Stakeholder handoff, customer-facing technical support, teaching, technical documentation

Professional Experience

Auckland Transport — **Nov 2025 – Feb 2026**
Computer Vision Intern (Full-time) — *Auckland, New Zealand*

- Designed and developed an internal operations dashboard enabling transport teams to query, process, and analyse operational data, improving workflow visibility and supporting day-to-day operational decisions.
- Tested AI vision models against real-world transport data and documented validation findings, helping teams understand where model outputs could and could not be trusted for decision-making.
- Worked in an Agile delivery environment, documenting technical issues and deployment behaviour to support reliable handoff between software, data, and operational stakeholders.

Cloudcell — **May 2025 – Oct 2025**
Software Development Intern — *Auckland, New Zealand*

- Conducted market research on affordable drug alternatives and translated findings into the design of a data platform that helped users search, compare, and analyse large-scale chemical compound data.
- Developed a drug similarity search platform using FastAPI, DuckDB, RDKit, and React/TypeScript. Processed approximately 2.47 million compound records from ChEMBL 35, built a local drug structure database, and precomputed Morgan fingerprints and molecular physicochemical properties. Supported SMILES/drug name search, Tanimoto/Cosine similarity search, molecular property filtering, structure visualization, and result export.
- Integrated the seyonec/ChemBERTa-zinc-base-v1 model to generate 768-dimensional molecular embeddings, enabling AI-powered molecular similarity search based on semantic representations. Designed a candidate drug post-processing workflow, including composite ranking, clustering, and drug pricing data aggregation, to support the discovery of potential low-cost drug alternatives.
- Delivered user-facing functionality iteratively through structured testing and documentation.
- Demo: cccccjin.github.io/Affordable-Drug-Alternatives

University of Auckland — **Mar 2025 – Jun 2025**
Graduate Teaching Assistant (Part-time) — *Auckland, New Zealand*

- Supported MECHENG 313 labs and tutorials involving explaining embedded systems, troubleshooting steps, and practical engineering procedures clearly to students with different technical backgrounds.

Axcelis Technologies, Inc. — **Jun 2023 – Jun 2024**
Field Service Engineer — *Shenzhen, China*

- Provided on-site installation, diagnosis, troubleshooting, maintenance, and repair support for complex semiconductor equipment and systems, testing and certifying equipment operating performance and system quality.
- Delivered customer training on system operation and maintenance, supporting reliable equipment handover to client teams.
- Owned field service activities spanning hardware, software, PC, and network system support.

Nikon Precision Shanghai Co., Ltd — **Jul 2019 – Aug 2022**
Field Service Engineer — *Shenzhen, China*

- Diagnosed complex equipment faults by analysing system logs and waveform data, applying Fourier Transform analysis and designing feedforward, notch, and peak filters to optimise system performance.
- Identified root causes of precision overlay anomalies using a 3σ statistical model of linear and nonlinear compensation values, comparing interferometer brightness and temperature variations across time periods.
- Calibrated laser heads and optical paths using a collimator and Zygo DynaFiz, and collected waveform logs via a Keyence NR-500 multi-channel data acquisition system to analyse vibration causes.

Research & Projects

SO-ARM101 **Mar 2026 – Now**
Independent R&D Engineer *Auckland, New Zealand*
Tech Stack: LeRobot, ACT, VLA (Pi0), PyTorch, Imitation Learning (IL), Reinforcement Learning (RL), Leader-Follower Teleoperation

- I built a data collection pipeline using a leader-follower robot arm, and collected 50 demonstration recordings across different tasks.
- I fine-tuned a Pi0 vision-language-action model using LoRA, and compared how it performed with and without updating the vision part of the model.
- I tried using reinforcement learning to improve the trained policy directly on the real robot.
- I trained a 300-million-parameter language model completely from scratch, to understand how these AI models are actually built.

Emotion-Aware Assistive Robotic Arm **Jan 2025 – Feb 2026**
120 Points Research *Auckland, New Zealand*

- Developed a ROS/MoveIt-based UR5e pick-and-place control system with joint-space and Cartesian trajectory execution, FK computation, path planning, and simulation validation.
- Integrated UR5e, Robotiq 85 gripper, and conveyor belt ROS interfaces for real-robot control, controller switching, parameter tuning, and Sim-to-Real deployment.
- Built an emotion-aware human-robot collaboration system that used facial expression recognition to adapt UR5e speed and interaction distance.
- Designed controlled experiments on robot speed and distance, and evaluated user experience using GodSpeed, NASA-TLX, and task performance metrics.
- Built an R-based analysis pipeline to clean experiment data, compute performance/workload metrics, and run repeated-measures ANOVA, paired t-tests, EMM, and FDR-corrected correlation analysis.
- Automated statistical tables and visualizations to identify how distance and task type affected task time and robot perception.

Emotion-Driven Interactive Education Chatbot **Jul 2024 – Nov 2024**
Frontend development (university project) *Auckland, New Zealand*

- Built a real-time emotion detection chatbot using OpenCV for video capture, YOLOv8 for face detection, and AlexNet (PyTorch) for emotion recognition.
- Implemented GUI for live video display with emotion labels at 20ms refresh intervals.

Education

University of Auckland **Jul 2024 – Jun 2026**
Master of Mechatronics Engineering (Graduated) *Auckland, New Zealand*

- Main Courses: Human-Robot Interaction (A), Robotics and Society (A), Multivariable Control System, Advanced Control.
- Thesis: An Experimental Investigation of Robot Speed, Proximity, and Task Pacing in Industrial Human–Robot Collaborative Assembly (A-)

Beijing Institute of Technology **Sep 2015 – May 2019**
Bachelor of Mechatronics Engineering *China*

- Main Courses: Mechanical Design and Theory, Programming C, Digital and Analog Electricity, Control Engineering, Theory and Applications of Embedded Systems, Machine Vision Technology. GPA: 2.82 (Top 30%)